

Contestant Number: _____

Time: _____

Rank: _____



JAVA PROGRAMMING (340)

STATE – 2020

Production Portion:

Program 1: Golf Course Management _____ (350 points)

TOTAL POINTS _____ (*350 points*)

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

No more than ten (10) minutes orientation
No more than ninety (90) minutes testing time
No more than ten (10) minutes wrap-up

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Workplace Skills Assessment Program competition.

You will have 90 minutes to complete your work.

Your name and/or school name should *not* appear on any work you submit for grading. Write your contestant number in the provided space. Staple all pages in order before you turn in your test.

To submit your project:

1. Create a folder on the flash drive using your contestant number as the name of the folder.
2. Copy your entire solution/project into this folder, including source code.
3. Ensure all the files required to run your program are present on the flash drive and will run on the flash drive.

Use the default package for the Java projects you create.

Golf Course Management

You have been hired by a local golf course to write a program to manage the golfer's data. All the data will be stored on a text file (provided) named "Players.txt". The file will contain a comma delimited line for each player that will provide their name and a series of scores. The programmer can assume that each record has a full name and at least one score. The file will be written in the format shown in Figure 1.

fName lName,##,##,##,##,##,##...

Figure 1

You will need to create a class named "player" based on the UML diagram provided below (figure 2). Note that the class will only have one overloaded constructor and no default constructor. The constructor will take in the comma delimited string read from the input file and parse out the name and fill the array/array list with the scores. The administrators of the course have decided to calculate the player's handicaps based on 90% of the average of the player's top five rounds. If the player has fewer than 5 rounds, their handicap will base on 90% of the average of all of the rounds played.

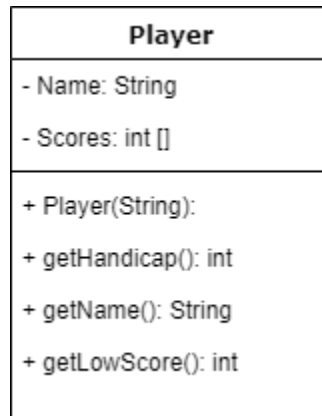


Figure 2

The main class should be name "PlayerDriver".On startup, the program should read the input file and populate a data structure containing all the player objects. The main program will need to use a menu feature that looks similar to the one below (Figure 3):

- | |
|--|
| <ul style="list-style-type: none">1- Find player handicap2- List all players best round3- List all players handicaps0- Exit |
|--|

Figure 3

All user input must be user validated. Option 1 will prompt the user to enter a name of a player. The program will search the data structure for that player. If the player is found, the program will list Name and Handicap. If the name is not found, the program should display a message notifying the user that the player was not found. Options 2 will list all the player's names and their best (lowest) scores. Option 3 will list all the players' names and their calculated handicaps. After options 1-3 are selected, the program should ask the user to press the enter key to continue.

Your program will be graded on the following criteria:

Solution and Project

The project is present on the flash drive.	_____	10 points
The project's main class is called Decoder.	_____	10 points

Program Execution

The program runs from the USB flash drive.	_____	10 points
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If the program does not execute, then the remaining items in this section receive a score of zero.

Menu is implemented.	_____	20 points
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The menu and its input is user validated	_____	10 points
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The program prompts for a player's name when option 1 is selected	_____	20 points
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The program prints the players name and handicap for option 1	_____	20 points
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The program lists all the players' lowest scores when option 2 is selected	_____	30 points
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The program lists all the players' handicaps when option 3 is selected	_____	30 points
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The program exits when option 0 is selected	_____	10 points
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Source Code Review

The contestant number is in a comment above the start of the main class.	_____	10 points
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The name of the main class is PlayerDriver.	_____	10 points
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The program has a class named Player	_____	20 points
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The Player class has a single overloaded constructor that takes in comma delimited string and parses it correctly	_____	30 points
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The Player class has a method getHandicap that calculates the handicap _____ 30 points

The Player class has methods getName and getLowScore that work
correctly (15 points each) _____ 30 points

The program uses a try-catch to handle any input file errors _____ 30 points

Code is commented throughout the program _____ 20 points

Total Points = _____ / 330 points